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Kristian Tyn Kai Chung

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My research ranges from frustrated magnetism to lattice gauge theory and generalized symmetries. On the frustrated magnetism side my current research efforts focus on the construction, classification, and realization of exotic spin liquids and topological orders exhibiting fractonic behavior, where excitations have constrained mobility due to carrying non-trivial crystalline symmetry multipoles. On the generalized symmetries side I study both modulated symmetries, where internal and spatial symmetries mix together, and non-invertible symmetries described by fusion categories rather than groups—how they can be broken and how they organize the excitation spectrum. Lattice gauge theory serves as a bridge between the two ends of my research, bridging from the realm of spin liquidity to the realm of theoretical models harboring generalized symmetries. At both ends of this research spectrum I am collaborating with experimental groups to find material (rare earth magnets) and synthetic platforms (such as Rydberg quantum simulators) where these exotic theoretical predictions can be observed and directly manipulated.

Academic Experience

Carl and Lillian Illig Postdoctoral Fellow — *Smalley-Curl Institute, Rice University* (2025 – now)

Postdoctoral Fellow — *Max Planck Institute for the Physics of Complex Systems* (2023 – 2025)

Ph.D. Physics — *University of Waterloo* (2017 – 2023)

M.Sc. Physics — *San Francisco State University* (2014 – 2017)

B.Sc. Physics — *University of the Pacific* (2008 – 2013)

Primary References

1. Michel J. P. Gingras — *University of Waterloo*
◇ Position: Professor, Canada Research Chair in Condensed Matter Physics & Statistical Mechanics
◇ Email: gingras@uwaterloo.ca
◇ Website: https://uwaterloo.ca/physics-astronomy/profile/gingras

2. Andriy Nevidomskyy — *Rice University*
◇ Position: Professor, Physics and Astronomy
◇ Email: nevidomskyy@rice.edu
◇ Website: https://profiles.rice.edu/faculty/andriy-nevidomskyy

3. Sergej Moroz — *Karlstad University*
◇ Position: Associate Professor of Physics
◇ Email: sergej.moroz@kau.se
◇ Website: https://www.kau.se/en/researchers/sergej-moroz

Additional References

4. Paul McClarty — *Laboratoire Léon Brillouin Saclay*
◇ Position: CNRS Researcher (DR2)
◇ Email: paul.mcclarty@cea.fr
◇ Website: https://scholar.google.com/citations?user=VRBh43gAAAAJ

5. Chistopher Hooley — *Coventry University*
◇ Position: Professor, Group Lead: Statistical Physics Group
◇ Email: ae4620@coventry.ac.uk
◇ Website: https://pureportal.coventry.ac.uk/en/persons/chris-hooley

6. Johannes Reuther — *Freie Universität Berlin*
◇ Position: Professor
◇ Email: johannes.reuther@fu-berlin.de
◇ Website: https://www.physik.fu-berlin.de/einrichtungen/ag/ag-reuther/index.html

Publications

Fracton Spin Liquid and Exotic Frustrated Phases in Ising-like Octochlore Magnets (2026)
◇ M. Stern, M. Burke, J. Romhanyi, M. Gingras, **K. Chung**
◇ arXiv:2603.12313 (in review at PRX)

Spontaneously Broken Non-Invertible Symmetries in Transverse-Field Ising Qudit Chains (2026)
◇ **K. Chung**, U. Borla, A. Nevidomskyy, S. Moroz
◇ Physical Review Letters 136, 216601 (2026)

Geometrically Frustrated Quadrupoles on the Pyrochlore Lattice and Generalized Spin Liquids (2026)
◇ **K. Chung**, S. Petit, J. Robert, P. McClarty
◇ Physical Review X **16**, 031065 (2026)

Mapping the Phase Diagram of a Frustrated Magnet (2026)
◇ **K. Chung**
◇ Physical Review B **114**, 084416 (2026)

Higgs Phases and Boundary Criticality (2024)
◇ **K. Chung**, R. Flores-Calderon, R. Carreira, P. Riberiro, S. Moroz, P. McClarty
◇ SciPost Physics **19**, 105 (2025).

2-Form U(1) Spin Liquids: Classical Model and Quantum Aspects (2023)
◇ **K. Chung**, M. Gingras
◇ Physical Review B **111**, 064417 (2025).

Probing Flat Band Physics in Spin Ice Systems via Polarized Neutron Scattering (2022)
◇ **K. Chung**, J. Goh, A. Mukherjee, W. Jin, D. Lozano-Gómez, M. Gingras
◇ Physical Review Letters **128**, 107201 (2022).

Coulomb Flux Tube on the Lattice (2017)
◇ **K. Chung**, J. Greensite
◇ Physical Review D **96**, 034512 (2017).

Activities

Generalized Symmetries Journal Club (Founder & Organizer) *MPI-PKS* (2023 – 2025)
◇ A journal club dedicated to exploring topics in generalized symmetries.
◇ Website: generalized-symmetries-journal-club.github.io.

Quantum Matters Grad Talks (Cofounder & Organizer) *University of Waterloo* (2019 – 2022)
◇ A condensed matter seminar series by and for graduate students to discuss their research.
◇ Website: sites.google.com/view/qmgradtalks.

Awards

Best Poster Award *International Center for Theoretical Physics (ICTP)* (2023)
◇ Venue: Conference on Fractionalization and Emergent Gauge Fields in Quantum Matter
◇ Title: 2-Form U(1) Spin Liquids: Classical Model and Quantum Aspects

W.B. Pearson Medal *University of Waterloo* (2022)
◇ Description: “In recognition of creative research as presented in the student’s thesis.”

Robert W. Maxwell Memorial Scholarship *San Francisco State University* (2016)

Additional Education

International Summer School on Computational Quantum Materials *University of Sherbrooke* (2022)

Online School on Ultra Quantum Matter *Perimeter Institute for Theoretical Physics* (2020)